

# 11 PCA

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$$1b) a) s^2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$X_1(5, 15, 15, 25) \quad \bar{x} = \frac{5 + 15 + 15 + 25}{4} = \frac{60}{4} = 15$$

$$s_1^2 = \frac{(5-15)^2 + (15-15)^2 + (15-15)^2 + (25-15)^2}{4}$$

$$= \frac{100 + 0 + 0 + 100}{4} = \underline{\underline{50}}$$

$$X_2(10, 30, 50, 30) \quad \bar{x} = \frac{10 + 30 + 50 + 30}{4} = \frac{120}{4} = 30$$

$$s_2^2 = \frac{(10-30)^2 + (30-30)^2 + (50-30)^2 + (30-30)^2}{4}$$

$$= \frac{400 + 0 + 400 + 0}{4} = \underline{\underline{200}}$$

$$X_3(5, 15, 15, 5) \quad \bar{x} = \frac{5 + 15 + 15 + 5}{4} = \frac{40}{4} = 10$$

$$s_3^2 = \frac{(5-10)^2 + (15-10)^2 + (15-10)^2 + (5-10)^2}{4}$$

$$= \frac{25 + 25 + 25 + 25}{4} = \frac{100}{4} = \underline{\underline{25}}$$

$$b) \text{ total} = 50 + 200 + 25 = \underline{\underline{275}}$$

$$c) PL_1 = (0.6, 0.8, 0)$$

Check if legal  $w_1^2 + w_2^2 + \dots = 1$

$$0.6^2 + 0.8^2 + 0^2 = 1 \quad 0.36 + 0.64 = 1$$

Projection =  $X_1 W_1 + X_2 W_2 + \dots$

$$D_1 = 5 \cdot 0.6 + 10 \cdot 0.8 + 5 \cdot 0 = 3 + 8 + 0 = 11$$

$$D_2 = 15 \cdot 0.6 + 30 \cdot 0.8 + 15 \cdot 0 = 9 + 24 + 0 = 33$$

$$D_3 = 15 \cdot 0.6 + 50 \cdot 0.8 + 15 \cdot 0 = 9 + 40 + 0 = 49$$

$$D_4 = 25 \cdot 0.6 + 30 \cdot 0.8 + 5 \cdot 0 = 15 + 24 + 0 = 39$$

$$PL_1 = \frac{11 + 33 + 49 + 39}{4} = \frac{132}{4} = 33$$

$$\text{Var}(PL_1) = \frac{(11-33)^2 + (33-33)^2 + (49-33)^2 + (39-33)^2}{4}$$

$$= \frac{484 + 0 + 256 + 36}{4} = \frac{776}{4} = 194$$

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$$\text{ratio} = \frac{\text{Var of component}}{\text{total Var}}$$

$$\text{ratio} = \frac{144}{275} \approx 0.705$$

PC<sub>1</sub> captures about 70.5% of total variance in the data set